FUSION-SUFFICIENT MOMENT EXCHANGE FOR DISTRIBUTED PROJECTED KLA-LMB TRACKING

Anonymous ICASSP Submission

ABSTRACT

Distributed labeled multi-Bernoulli (LMB) trackers may transmit full Gaussian-mixture (GM) fusion messages, although the receiver studied here projects each label before Gaussian Kullback–Leibler average (KLA) fusion. We use this receiver factorization to define the message interface. Let $\mathcal{P}$ denote label-wise moment projection and $\mathcal{G}_\omega$ the projected Gaussian fusion operator. Idempotence of $\mathcal{P}$ makes the receiver $\mathcal{F}_\omega = \mathcal{G}_\omega \circ \mathcal{P}$ satisfy $\mathcal{F}_\omega = \mathcal{F}_\omega \circ \mathcal{P}$. Sender-side projection therefore preserves the fusion output under matched source-label presence, weights, delivery masks, and numerical conventions. We implement a typed binary fusion-sufficient moment message and compare it with full GM fusion messages under the same codec. Across 50 paired eight-sensor trials, moment exchange reduces attempted application-layer bytes by 58.28% (95% bootstrap interval: 57.92–58.64%) and delivered bytes by 58.27%. All 1,119,037 audited label-instance fusion outputs match exactly, as do all tracking metrics. This is an interface result for the specified single-round projected receiver, not a general equivalence of Gaussian-mixture KLA densities.

***Index Terms*—** Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, fusion-sufficient message

1. INTRODUCTION

Distributed multi-object tracking lets a sensor network combine complementary views without centralizing all measurements. Labeled random finite-set filters, especially labeled multi-Bernoulli (LMB) filters, are well suited to this task because they represent target existence, state uncertainty, and identity in one posterior [1, 2, 3]. Their communication cost can be high: each transmitted LMB posterior may contain many labels and multiple Gaussian components per label.

Communication-aware RFS trackers commonly reduce transmission frequency, select mixture components, or project state estimates before exchange [4, 5, 6, 7, 8]. These methods change when estimates are sent, which mixture terms are shared, or how compact estimates are formed. Our question is different: for a fixed receiver, which transmitted fields can affect its output at all?

We answer that question for a single-round projected Gaussian KLA-LMB receiver. Its implementation aligns labels, projects each Gaussian-mixture Bernoulli to existence, mean, and covariance, and only then performs canonical Gaussian fusion. Writing the receiver as $\mathcal{F}_\omega = \mathcal{G}_\omega \circ \mathcal{P}$ exposes an operator identity: idempotence of $\mathcal{P}$ gives $\mathcal{F}_\omega = \mathcal{F}_\omega \circ \mathcal{P}$. Hence the sender may transmit $\mathcal{P}(\pi)$ instead of the full mixture without changing this receiver's fusion output. Figure 1 illustrates the commuting paths.

Our contributions are threefold. First, we formalize this receiver-induced message interface and state the exact conditions under which sender-side moment projection preserves the fusion output. Second, we implement a typed binary codec and measure the resulting full GM and fusion-sufficient moment messages at the same application-layer boundary. Third, a frozen 50-trial paired evaluation shows a 58.28% mean reduction in attempted bytes, with zero residual across 1,119,037 label-level fusion comparisons. We claim neither equality of the underlying mixture densities nor sufficiency for a mixture-aware or multi-round receiver.

2. RELATED WORK

Random finite-set filters provide a Bayesian framework for tracking uncertain target cardinality and measurement association. PHD filters introduced a tractable first-moment recursion [9]; labeled RFS models add persistent identities [1]; and LMB filters provide a practical independent-Bernoulli approximation [2]. Multi-sensor GLMB/LMB variants extend this framework to cooperative sensing [3]. When cross-correlations are unknown, KLA/GCI fusion combines densities through a normalized weighted geometric mean [10]; RFS instances include exponential-mixture PHD and multi-Bernoulli fusion [11, 12]. We do not change the local LMB recursion or propose a new KLA rule. We analyze the representation consumed by one projected Gaussian receiver.

Communication is reduced at several layers. Event-triggered LMB methods change when full updates are sent [4, 5, 6]; related triggers operate on $M\delta$-GLMB labels or hypotheses [13]. Partial-consensus PHD fusion selects Gaussian components [7], whereas state and covariance projection sends compact estimates at a common time [8]. Our design is different in origin: the fixed receiver operator induces the message fields. The resulting equivalence is conditional on that operator and therefore narrower than general Gaussian-

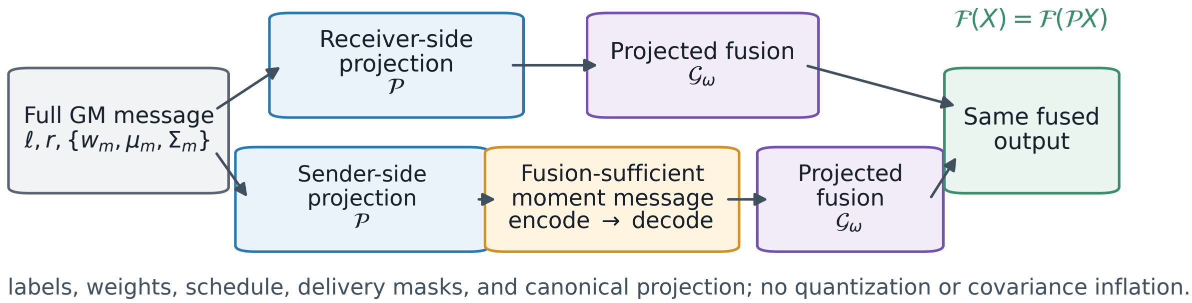

Fig. 1. Receiver-induced message design. The full-message path first applies $\mathcal{P}$ at the receiver. Idempotence lets the same projection move to the sender, so both paths present identical label-wise moments to $\mathcal{G}_\omega$ and produce exactly the same fused LMB output under the stated conditions.

mixture KLA approximation.

3. FUSION-SUFFICIENT MOMENT EXCHANGE

3.1. Label-Wise Moment Projection

At time k, source j maintains an LMB posterior

$$\pi_j = \{(r_j^\ell, p_j^\ell) : \ell \in \mathcal{L}_j\},$$
$$p_j^\ell(x) = \sum_{m=1}^{M_j^\ell} w_{jm}^\ell \mathcal{N}(x; m_{jm}^\ell, P_{jm}^\ell). \tag{1}$$

For normalized w_{jm}^ℓ, define the label-wise projection $\mathcal{P}$ by

$$\mathcal{P}(\pi_j) = \{(r_j^\ell, \mathcal{N}(\mu_j^\ell, \Sigma_j^\ell))\}_\ell, \quad \mu_j^\ell = \sum_m w_{jm}^\ell m_{jm}^\ell, \tag{2}$$

$$\Sigma_j^\ell = \sum_m w_{jm}^\ell \left[P_{jm}^\ell + (m_{jm}^\ell - \mu_j^\ell)(m_{jm}^\ell - \mu_j^\ell)^\top\right]. \tag{3}$$

This keeps the label and existence probability and replaces the mixture by the Gaussian with identical first two moments. The projection is idempotent: $\mathcal{P}(\mathcal{P}(\pi)) = \mathcal{P}(\pi)$.

3.2. The Projected Gaussian KLA-LMB Receiver

Receiver i first applies $\mathcal{P}$ to every delivered source, aligns the union of labels, and then fuses one Gaussian per label. Let $\mathcal{S}_i^\ell$ contain self and delivered neighbors carrying label ℓ. Let α_{ij} and β_{ij} be the spatial and existence weights. Missing-label renormalization gives

$$a_{ij}^\ell = \frac{\alpha_{ij}}{\sum_{q\in\mathcal{S}_i^\ell} \alpha_{iq}}, \qquad b_{ij}^\ell = \frac{\beta_{ij}}{\sum_{q\in\mathcal{S}_i^\ell} \beta_{iq}}. \tag{4}$$

The Gaussian canonical parameters are

$$K_i^\ell = \sum_{j\in\mathcal{S}_i^\ell} a_{ij}^\ell (\Sigma_j^\ell)^{-1}, \quad h_i^\ell = \sum_j a_{ij}^\ell (\Sigma_j^\ell)^{-1} \mu_j^\ell,$$
$$\bar{\Sigma}_i^\ell = (K_i^\ell)^{-1}, \qquad \bar{\mu}_i^\ell = \bar{\Sigma}_i^\ell h_i^\ell. \tag{5}$$

Define the Gaussian overlap by

$$\eta_i^\ell = \int \varphi_i^\ell(x)\, dx,$$
$$\varphi_i^\ell(x) = \prod_{j\in\mathcal{S}_i^\ell} \mathcal{N}(x; \mu_j^\ell, \Sigma_j^\ell)^{a_{ij}^\ell}. \tag{6}$$

The Bernoulli terms are

$$q_1^\ell = \eta_i^\ell \prod_j (r_j^\ell)^{b_{ij}^\ell}, \quad q_0^\ell = \prod_j (1 - r_j^\ell)^{b_{ij}^\ell}, \quad \bar{r}_i^\ell = \frac{q_1^\ell}{q_0^\ell + q_1^\ell}. \tag{7}$$

where all products in Eq. (7) are over $j \in \mathcal{S}_i^\ell$. The implementation clamps each r_j^ℓ to $[10^{-9}, 1 - 10^{-9}]$ and lower-bounds $q_0^\ell + q_1^\ell$ by machine epsilon; both paths use the same guards. Denote Eqs. (5)–(7) by $\mathcal{G}_\omega$, where $\omega = (\alpha, \beta)$, and the receiver by $\mathcal{F}_\omega = \mathcal{G}_\omega \circ \mathcal{P}$.

Proposition 1 (fusion-output equivalence). For each source, assume identical active-label presence after selection and pruning; identical message-type-independent spatial and existence weights; identical delivery masks; common projection and covariance regularization; disabled covariance inflation; and a common lossless codec. Then

$$\mathcal{F}_\omega(\pi_1, \ldots, \pi_n) = \mathcal{F}_\omega(\mathcal{P}\pi_1, \ldots, \mathcal{P}\pi_n). \tag{8}$$

Proof. Projection preserves each source-label's r, μ, and Σ and does not change its presence in $\mathcal{S}_i^\ell$. Hence the renormalized a_{ij}^ℓ, b_{ij}^ℓ, canonical K_i^ℓ, h_i^ℓ, overlap η_i^ℓ, and Bernoulli terms q_0^ℓ, q_1^ℓ are unchanged. The common codec and covariance regularizer therefore produce the same $\bar{r}_i^\ell, \bar{\mu}_i^\ell, \bar{\Sigma}_i^\ell$. Equivalently, $\mathcal{G}_\omega \circ \mathcal{P} \circ \mathcal{P} = \mathcal{G}_\omega \circ \mathcal{P}$ by idempotence. □

We call $\mathcal{P}(\pi)$ *fusion-sufficient* only with respect to this $\mathcal{F}_\omega$: messages in the same equivalence class yield the same fusion output. Equation (8) does not imply equal posterior densities. It need not hold if a receiver preserves mixture modes, adapts weights from higher-order structure, uses different projection regularization, or reuses a compressed result in a multi-round algorithm.

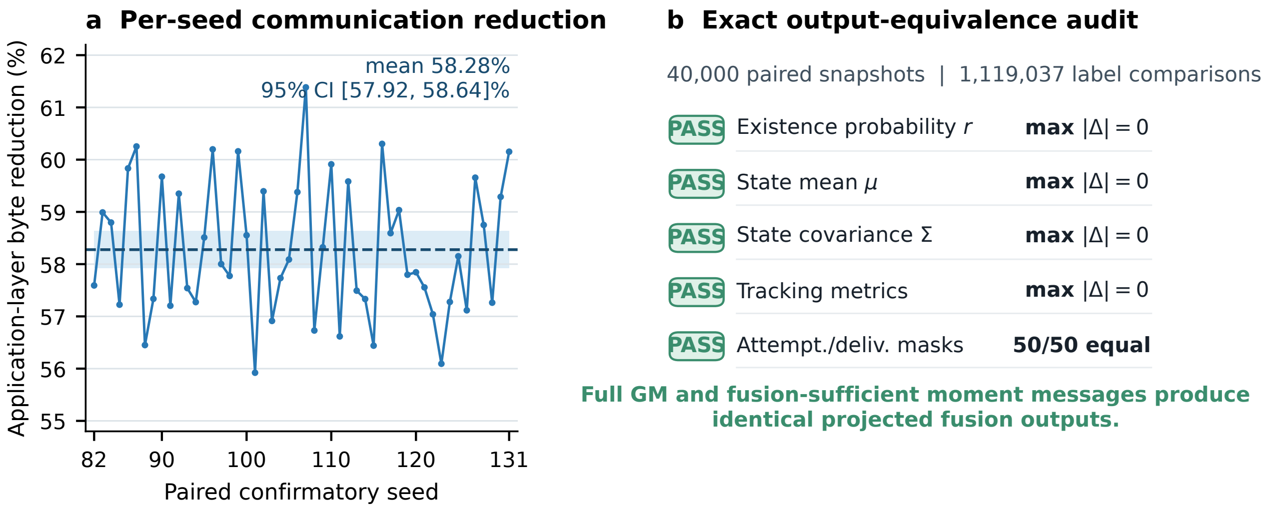

Fig. 2. Confirmatory evidence. Left: every paired seed reduces attempted application-layer bytes; the line and band show the 58.28% mean and 95% bootstrap interval. Right: all 40,000 paired sensor-time snapshot comparisons and 1,119,037 matched label-instance comparisons have identical labels, r, μ, and Σ, producing zero tracking-metric deltas.

3.3. Typed Message and Byte Accounting

Both arms use the same versioned, little-endian binary codec: a 24-byte message header, a 20-byte label header, and binary64 component fields for weight, mean, and the upper covariance triangle. For state dimension d, a message with M_ℓ components per label has

$$B = 24+\sum_{\ell}\left[20+8M_\ell\left(1+d+\frac{d(d+1)}{2}\right)\right] \text{ bytes.} \quad (9)$$

The full message retains every M_ℓ in Eq. (1); the proposed message sets $M_\ell = 1$ using Eqs. (2)–(3). Attempted bytes are counted after encoding and before the drop draw; delivered bytes count messages decoded after successful delivery. No other tracker variable, scheduling decision, or fusion parameter changes.

4. EXPERIMENTS

4.1. Paired Confirmatory Protocol

We evaluate on an eight-sensor 4+4 formation benchmark with 100 simulation steps per trial. Each node runs the same local LMB prediction and measurement update, then performs one synchronous fusion round. The fixed graph has 16 undirected edges (two four-node cliques and four cross-clique links). Detection probability is 0.9, mean clutter count is 3, and measurement-noise standard deviation is 3. Node drop probabilities $[0, 0.1, 0.2, 0.5]$ have multiplicities $[1, 4, 1, 2]$ and are randomly assigned per trial, giving mean 0.2.

We compare only periodic full GM fusion messages and fusion-sufficient moment messages. The 50 paired trials use seeds 82–131. Within each pair, target trajectories, measurements, schedules, labels, fusion weights, and drop uniforms are identical; attempted and delivered masks are checked for equality. Local accuracy uses E-OSPA, while agreement uses consensus OSPA, position disagreement, and cardinality dispersion [14, 15]. The primary communication statistic is the mean of the 50 per-trial attempted-byte reductions. We report a paired percentile-bootstrap interval with 10,000 resamples and fixed bootstrap seed 20270710.

Table 1. Paired confirmatory means over 50 trials. Att./Del. MB are attempted/delivered application-layer megabytes per trial; Cons. is consensus OSPA. Lower is better.

Message	Att. MB	Del. MB	Local	Cons.
Full GM	25.084	20.091	2.076	1.951
Moment msg.	10.458	8.378	2.076	1.951

4.2. Results and Scope

Table 1 and Fig. 2 show a 58.277% mean per-trial reduction in attempted bytes (95% interval: 57.923–58.636%; minimum 55.922%). The mean delivered-byte reduction is 58.267%. Across all trials, full and moment messages attempt 1,254,185,200 and 522,888,880 bytes, respectively; delivered totals are 1,004,548,968 and 418,898,448 bytes.

The output audit is stronger than rounded metric agreement. All 40,000 paired sensor-time snapshot comparisons have identical label sets. Across 1,119,037 matched label-instance comparisons, the maximum residuals in existence probability, mean, and covariance are each exactly zero. Consequently, local E-OSPA, consensus OSPA, position disagreement (4.614), and cardinality dispersion (0.203) are identical in every paired trial. This matches Proposition 1 and also checks the codec boundary.

5. DISCUSSION

Proposition 1 is algebraically simple because it states a receiver contract, not a new approximation to Gaussian-mixture KLA. Its practical content depends on showing that the studied fusion path truly factors through $\mathcal{P}$. A comparison would be circular if both arms projected before transport. Our full arm instead encodes every original mixture component, and the receiver decodes the message before invoking $\mathcal{P}$. The moment arm invokes the same canonical projection at the sender and encodes one component. We compare decoded outputs at every sensor and time step. This crosses the serialization boundary and tests whether discarded mixture structure affects $\mathcal{F}_\omega$.

Equation (9) also explains why 58.28% is not a universal compression ratio. Let $c_d = 1 + d + d(d+1)/2$. For fixed labels, the saved payload is $8\sum_\ell (M_\ell - 1)c_d$ bytes, while both message and label headers remain. No bytes are saved when every label already has one Gaussian component. Savings increase with mixture multiplicity, but the header fraction limits the realized ratio for small messages. The confirmatory result therefore characterizes this tracker workload rather than an implementation-independent constant.

The equality can fail if an operation after $\mathcal{P}$ consumes discarded structure. Examples are component gating, label pruning based on modality, weights derived from mixtures, or later rounds that reuse individual modes. System details can also break commutation. Quantization before projection need not match quantization after projection, and different covariance regularizers can produce different canonical moments. Payload-dependent delivery or fragmentation can change masks, violating a condition of Proposition 1.

Our byte boundary contains the encoded application payload. It excludes MAC/PHY framing, network and transport headers, fragmentation, retransmission, and radio energy. The simulated loss probability is payload-size independent, which isolates representation size from delivery. This design supports a clean interface test, but hardware-level savings require packetization, quantization, and energy measurements on the target link.

The reusable idea is a design procedure rather than a claim of broad equivalence. Identify the receiver's first irreversible map, factor the consumer through that map, check whether it is canonical and idempotent, and promote its output to an explicit message type. Then validate equivalence after serialization under matched schedules and masks. Event triggering and component selection decide when or which estimates to send. Receiver-induced message design instead asks which fields a specified consumer can observe. Its guarantee is exact within that contract and intentionally silent outside it.

This distinction does not imply that transmitting means and covariances is itself new. State/covariance projection already supports asynchronous multi-radar fusion [8], and partial consensus can select Gaussian components [7]. Event-triggered LMB methods reduce full-update frequency [4]. Adaptive variants refine the trigger [5]. Our contribution is the receiver-side certificate that identifies an exact message equivalence class for one implemented consumer. The evidence standard changes accordingly. Since the schedule, masks, and fusion rule remain fixed, success is not a trade-off between accuracy and communication. It is equality of decoded fusion outputs, accompanied by a workload-specific byte reduction.

The evaluation has two main validity limits. First, the interval quantifies variation across trajectories and drop draws within one eight-sensor model. It does not quantify scenario generality. The sensors use homogeneous detection and measurement models, one graph, binary64 fields, and a single synchronous round. A different pruning rule can also change M_ℓ and therefore the byte ratio. Second, exact zero residual uses one canonical projection implementation on both paths. Independent implementations can differ in summation order or regularization, even when they represent the same real-valued moments. Proposition 1 predicts equality only after those conventions are matched. Broader validation should vary state dimension, mixture multiplicity, sensor heterogeneity, fusion weights, and packetization under frozen comparison gates.

6. CONCLUSION

We derived a fusion-sufficient moment message from a projected Gaussian KLA-LMB receiver. Moving the receiver's idempotent moment projection to the sender preserved every audited fusion output. It reduced attempted application-layer bytes by 58.28% across 50 paired trials. This exact result is receiver-specific, not a general mixture-aware KLA equivalence result. Future tests should cover quantized links, multi-round reuse, and receivers that retain multimodality.

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